# LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data

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## Objective, Dataset and Model

| Objective | Dataset | Model |
|---|---|---|
| Predict the outcome: 0 = No diabetes; 1 = Diabetes. | Pima Indians Diabetes Database from Kaggle; file: **diabetes.csv**. | Median imputation + StandardScaler + balanced Logistic Regression. |

## Dataset and Workflow

**Input features:** Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

**Workflow:** Load Kaggle CSV → inspect data → replace invalid zeros → train-test split → median imputation → scaling → train Logistic Regression → evaluate the model → interpret coefficients.

## Evaluation Metrics

| Metric | Result | Meaning |
|---|---|---|
| Accuracy | 0.734 | Overall correct predictions |
| Precision | 0.603 | Reliability of positive predictions |
| Recall | 0.704 | Actual positive cases detected |
| F1-score | 0.650 | Balance of precision and recall |
| ROC-AUC | 0.813 | Overall class-separation ability |

## Confusion Matrix

| | Predicted 0 | Predicted 1 |
|---|---|---|
| **Actual 0** | TN = 75 | FP = 25 |
| **Actual 1** | FN = 16 | TP = 38 |

## Key Findings

The model obtains a ROC-AUC of about 0.813 and detects approximately 70.4% of the positive cases in the fixed test split. Glucose and BMI have the strongest positive model coefficients. Recall is especially important in a screening-style classification task because false negatives represent missed positive cases.

## Conclusion

In this project, Logistic Regression was used to predict diabetes outcomes using the Pima Indians Diabetes Dataset. The data was prepared by replacing invalid zero values with missing values, filling missing values with the median, and standardizing the input features. The dataset was divided into training and testing sets, and a balanced Logistic Regression model was trained. Its performance was assessed using accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix. The reported results indicate that the model can identify a portion of positive cases and provide a useful demonstration of a machine-learning classification workflow. This project is intended for educational purposes only and must not be used as a medical diagnosis tool.